



Loop a Longer Pattern

Ontario Math C3.1 — repeating events

Name: _____

Date: _____

★ I can loop a pattern that has more than one action inside.

A pattern with more parts

A drum beat can be clap, clap, stomp — clap, clap, stomp. The whole group repeats.

A loop runs EVERY block inside it each time around. So a loop with three actions repeats all three together. Count how many of each Bit does.

Read this loop

when green flag clicked

repeat 4

clap

clap

stomp

Three actions inside (clap, clap, stomp) all run each repeat. 4 repeats = 12 actions in all.

one repeat = 3 actions

clap

clap

stomp

× 4



repeat 4 → 12 actions in all

The 3-action group is one repeat — the loop does it 4 times.

1 Count it

The loop is repeat 4 with clap, clap, stomp inside. How many actions in ONE repeat? _____ How many actions in TOTAL? _____

2 Predict, then check

Change repeat 4 to repeat 5. PREDICT: how many STOMPS does Bit do? How many CLAPS? Then check.

3 Write your own

Bit's pattern is wave, wave, bow. To do the whole pattern 4 times, fill in: repeat _____. How many WAVES is that in total? _____

4 Challenge

Bit's loop did 6 STOMPS. The pattern inside has 1 stomp each repeat. How many times did the loop repeat? How do you know?

Big idea

When a loop has more than one block inside, ALL of them run each repeat. To count one action, multiply how many per repeat by the number of repeats.



Loop a Longer Pattern

Quick facilitation notes & answer key

What this worksheet teaches

A loop runs every block inside it each time, so a multi-action pattern repeats as a unit. Students count each action across the repeats by multiplying, and work backwards from a total. No coding experience needed.

Before you start

No computer needed — the worksheet shows the blocks and what Bit does, so you can teach from the printout. Optional: a Scratch-style app to drag the blocks and watch Bit run — you only press the green flag.

How to lead it (about 30 minutes)

1. Show it (you do). The pattern clap, clap, stomp is 3 actions; 4 repeats make 12 actions.
2. Do one together. Exercise 1 — 3 actions per repeat, 12 in total.
3. Let them work. Students count each action type (Ex 2), write a repeat (Ex 3), and work backwards from a total (Ex 4).

Answer key (these are the verified correct answers)

Exercise 1 — 3 actions per repeat; 12 actions total. Exercise 2 — 5 stomps and 10 claps.

Exercise 3 — repeat 4; 8 waves total. Exercise 4 (challenge) — 6 repeats (6 stomps ÷ 1 stomp per repeat).

If students get stuck — and ways to adjust

Watch for: counting the whole pattern as one action. Count each kind of action separately.

Make it easier: count how many of one action are in a single repeat, then multiply.

Make it harder: ask the total claps for the pattern repeated 7 times.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The child multiplies each action by the repeats and works backwards from a total.